DHBW Ravensburg

Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | October 24, 2026 · *drifting stream*

LECTURE II

Introduction to Machine Learning Systems Design

This lecture in one slide

II Introduction to Machine Learning Systems Design

Business and ML Objectives
Requirements for ML Systems
Iterative Process
Framing ML Problems
Mind Versus Data

why this lecture matters, and what you can do after it

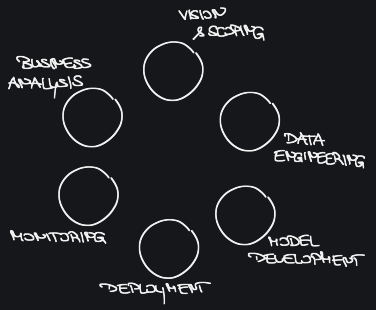
1

Business and ML Objectives

System Design

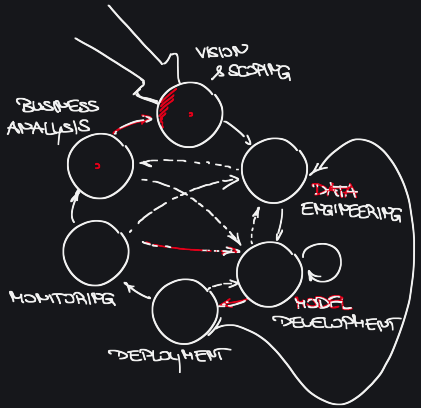
ML systems design takes a **system approach** to MLOps: the business requirements, the data stack, infrastructure, deployment, monitoring, and their stakeholders work together to satisfy the specified objectives and requirements.

If this system is built for a business, it must be driven by business objectives, which will need to be translated into ML objectives to guide the development of ML models.



[1]

It is not crazy, it is just complex



Objectives and Key Results



Objective what is to be achieved.

Key result how we get there: specific, time bound, measurable and verifiable.

One level's key result becomes the next level's objective, all the way down the org chart. The ML metric lands at the bottom.

[2]

“There is one and only one social responsibility of business … to increase its profits.”

Milton Friedman, 1970. While most companies want to convince you otherwise.

[3]

Your ML metric is a key result, very seldom the objective.

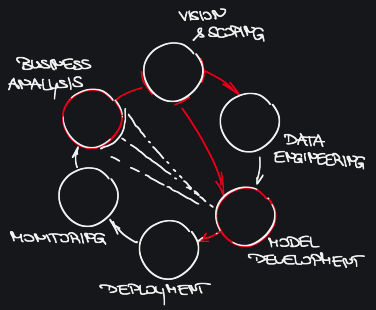
Most companies don’t care about increasing a model’s accuracy from 94% to 94.2% unless it moves some business metrics.

[1]

Tie ML to business

For an ML project to succeed within a business organization, it's crucial to tie the performance of an ML system to the overall business performance. At best the ML metric becomes a **proxy** for a business metric.

- **Ad click-through rate** every increase results in actual ad revenue.
- **Fraud detection** every fraudulent transaction stopped results in actual money saved.



Netflix take-rate

Netflix measures the performance of their recommender system using take-rate.

$$\text{take-rate} = \frac{\text{quality plays}}{\text{recommendations seen}}$$

The higher the take-rate, the better the recommender system.

Put in context with total streaming hours and the subscription cancellation rate: a higher take-rate also results in more streaming hours and fewer cancellations.

Over years of personalization, monthly churn fell by several percentage points, worth more than $1B per year [4].

[4, 5]

Who gains from the link

How ML metrics influence business metrics usually takes an experiment: run an **A/B test** and choose the model that leads to better business metrics, regardless of whether it has better ML metrics.

Manager

You know you pour money into a profitable initiative.

Team lead

Your ML team will not get killed with the project anytime soon.

Developer

Communicate it well and you will have a great time in performance reviews.

[1]

How important are ML systems for your company?

Rate it. Then compare it to your estimate for Netflix.

Which are the business key indicators of your company or division?

Which ML metrics can you design that are tied to them?

For all the wrong reasons

40%

of 2,830 European startups classed as AI companies showed no evidence of AI material to their value proposition.

Many companies like to say that they use ML because “being AI powered” alone already helps them attract customers, regardless of whether the AI part actually does anything useful.

Hype sells.

[6, 7]

2

Requirements for ML Systems

Forty-two

"I think the problem, to be quite honest with you, is that you've never actually known what the question is."

We can't say that we've successfully built an ML system without knowing what **requirements** the system has to satisfy.

They vary from use case to use case. Most systems should have four: reliability, scalability, maintainability, and adaptability.

Deep Thought: forty-two

[1, 8]

The four requirements

- **RELIABILITY**
  Correct function at desired level of performance in the face of adversity.
- **SCALABILITY**
  With respect to complexity, throughput, and other types of growth.
- **MAINTAINABILITY**
  Structure workload, infrastructure and tool landscape so that you can keep up.
- **ADAPTABILITY**
  Capacity to discover potentials for performance improvement and allowing updates without service interruptions.

Reliability

Definition: Reliability

The system should continue to perform the correct function at the desired level of performance even in the face of adversity: hardware or software faults, and even human error.

Traditional software warns you: a crash, a runtime error, a 404.

ML systems can fail silently. `model.predict()` is called correctly, but the predictions are wrong, and without ground truth labels nobody knows. Translate a sentence into a language you don't know.

[1, 9]

Reliability

Definition: Reliability

The system should continue to perform the correct function at the desired level of performance even in the face of adversity: hardware or software faults, and even human error.

Fail loudly. A fault surfaces immediately and visibly, where it happens, instead of carrying on with bad state.

Degrade gracefully. Under overload, lower the quality of the response rather than drop it: a smaller index, a cheaper ranking, a simpler model.

[10, 11]

Scalability

Whichever way your system grows, there should be **reasonable ways** of dealing with that growth.

Complexity

Last year a logistic regression on 1 GB of RAM. This year a 100 million parameter network that needs 16 GB.

Traffic volume

10,000 prediction requests daily at launch. Between 1 million and 10 million once the user base grows.

Model count

One model per enterprise customer. A startup Huyen worked with had 8,000 models in production.

[1, 9]

Scaling is not only resources

Resource scaling. At peak 100 GPUs, most of the time 10. Autoscaling scales up and down with usage, and is tricky to implement.

Artifact management. One model you can monitor and retrain by hand. With one hundred, monitoring and retraining need to be automated, and every model reproducible.

Even Amazon fell victim: autoscaling failed on Prime Day 2018 and the system crashed.

One hour of downtime was estimated to cost Amazon between $72 million and $99 million.

[1, 12]

Maintainability

ML engineers, DevOps engineers, and subject matter experts, from very different backgrounds, own different parts of the process.

- Everyone works with tools they are comfortable with
- Code is documented
- Code, data, and artifacts are **versioned**
- Models are reproducible when the original authors are not around
- Problems get solved without finger pointing

"Over time, many different people will work on the system … and they should all be able to work on it productively."

[1, 9]

Adaptability

Definition: Adaptability

To adapt to shifting data distributions and business requirements, the system should have some capacity for both discovering aspects for performance improvement and allowing updates without service interruption.

ML systems are part code, part data, and data can change quickly.

Lecture VIII, Data Distribution Shifts and Monitoring. Lecture IX, Continual Learning and Test in Production.

[1]

Alignment and Controllability

Alignment

Make AI systems behave in line with human intentions and values.

Controllability

Humans can supervise the system and intervene when it deviates from the expected behavior.

Robustness, Interpretability, **Controllability**, Ethicality: the four objectives of alignment.

LLMs and agents take over decision making, or at least augment it.

[13]

In the loop, on the loop

Controllability is a design decision: **who decides**, and who acts.

Human	Decides	Acts
in the loop	human, on a proposal	system
on the loop	system, human can veto	system
out of the loop	system	system

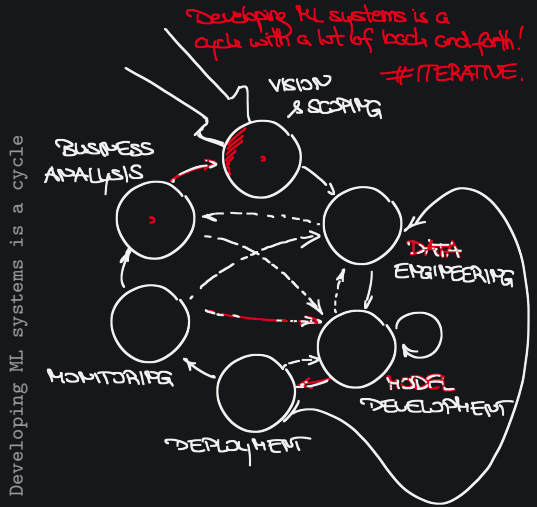
Served to a lot of users, scale, bias and performative prediction become requirements.

Interpretability and the human machine interface decide whether the human can intervene at all. Lecture I: human on the loop.

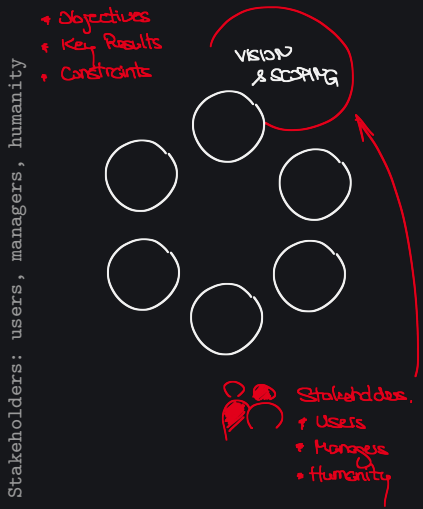
[14, 15]

3

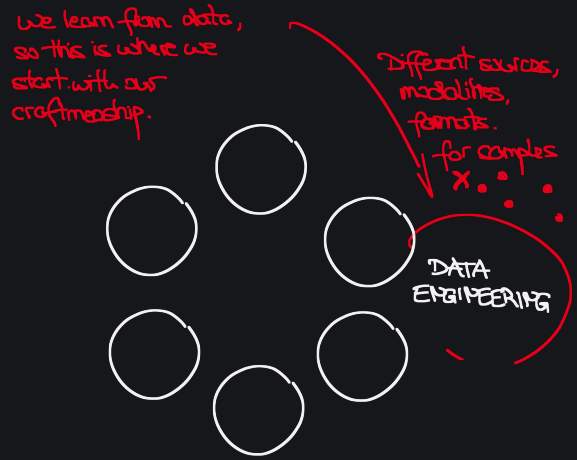
Iterative Process

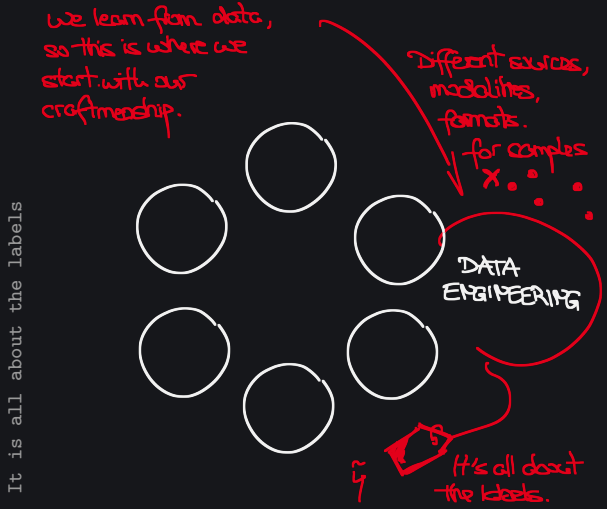

Developing ML systems is a cycle
Developing ML systems is a cycle with a lot of back and forth!
#ITERATIVE.
VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING

Vision and scoping

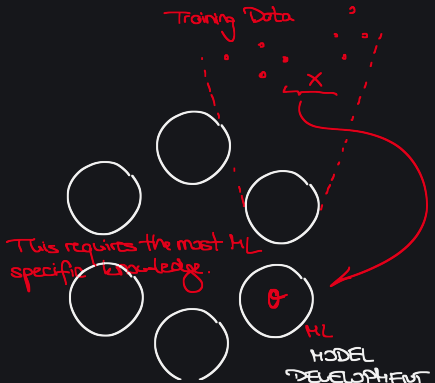

Stakeholders: users, managers, humanity
• Objectives
• Key Results
• Constraints
VISION & SCOPING
Stakeholders.
• Users
• Managers
• Humanity

Data engineering: the samples

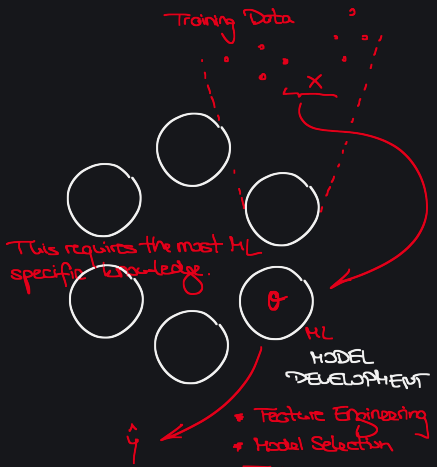



We learn from data, so this is where we start with our craftmanship.
Different sources, modalities, formats.
for complex
DATA ENGINEERING
It's all about the labels

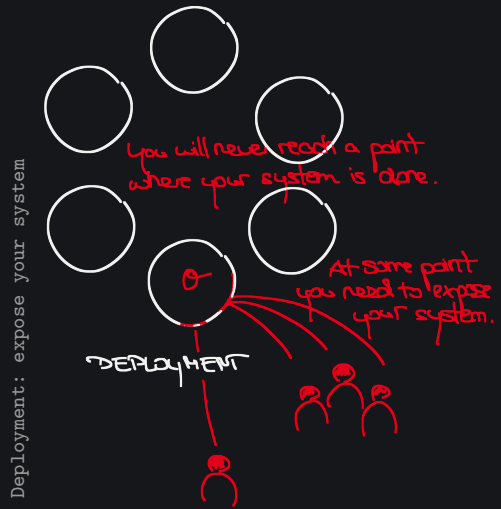
Model development



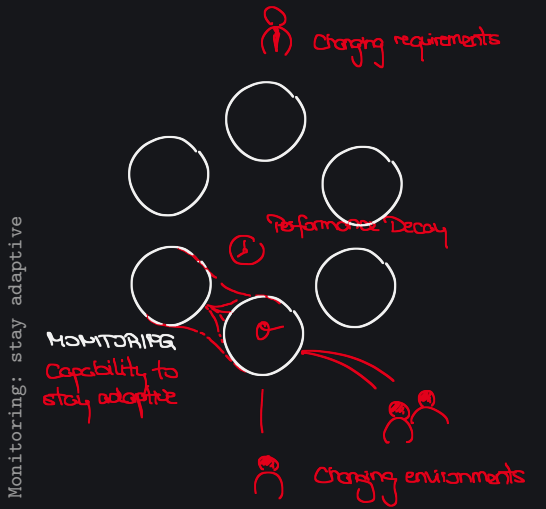
Feature engineering to evaluation



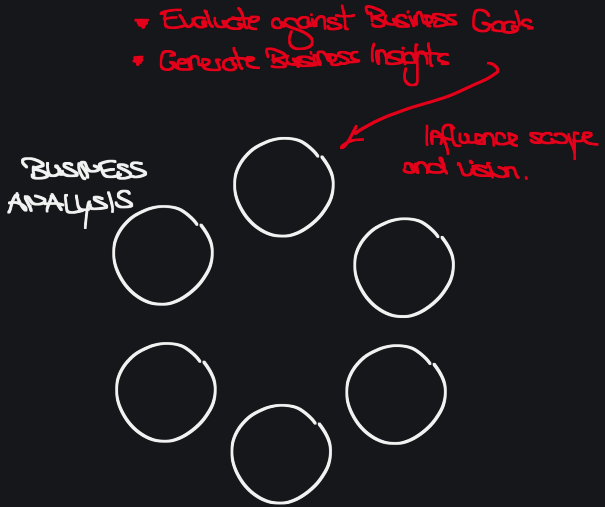
Deployment: expose your system



Monitoring: stay adaptive



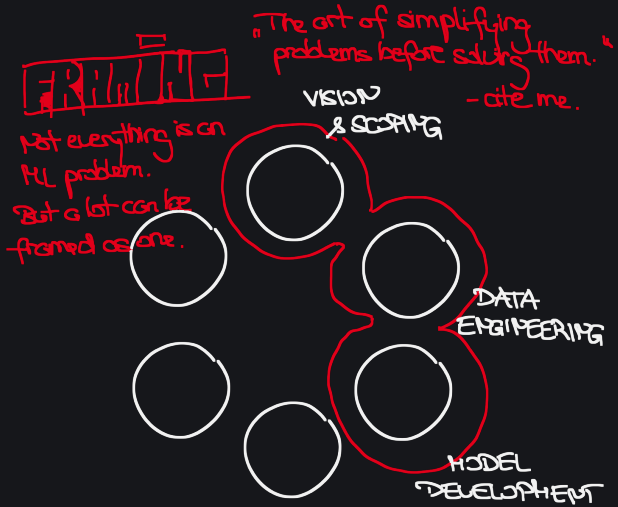
Business analysis closes the loop



4

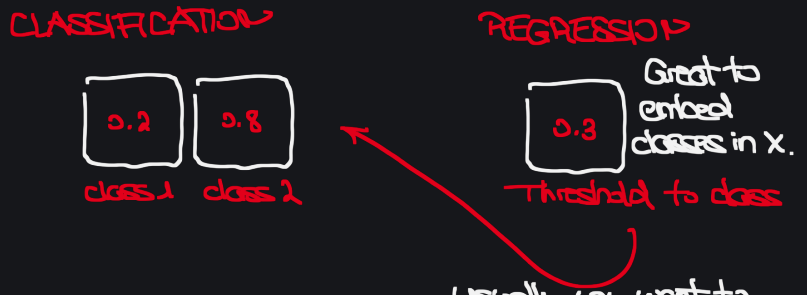
Framing ML Problems

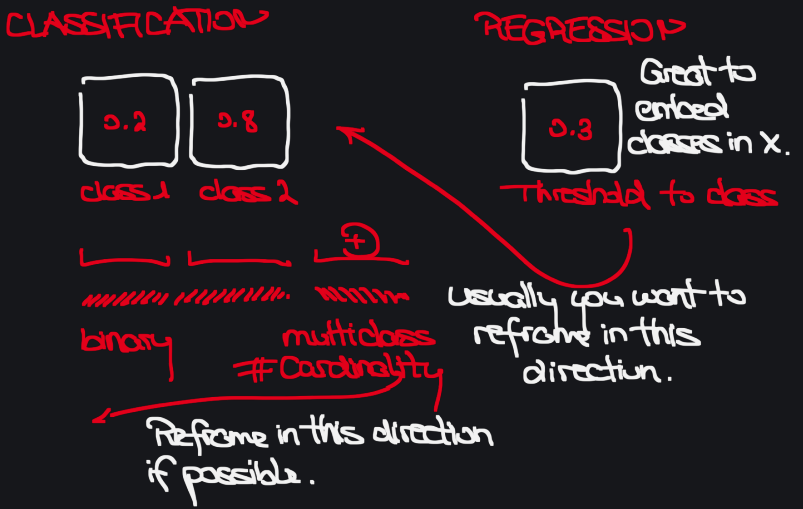
A lot can be framed as an ML problem



Types of ML Tasks

Classification versus regression




CLASSIFICATION
0.2
0.8
class 1
class 2
REGRESSION
0.3
Great to embed classes in X.
Threshold to class
binary
multiclass
#Cardinality
Usually you want to reframe in this direction.
Reframe in this direction if possible.

Multiclass versus multilabel classification

Multiclass

Multilabel

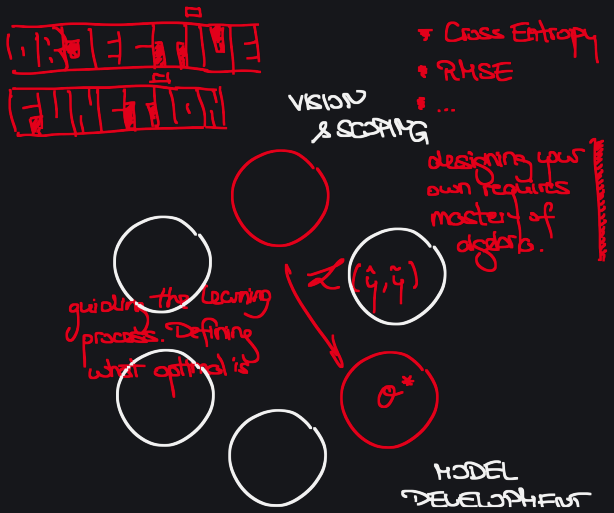
Multiple ways to frame a problem

!! Restrain from „buying" this with an increasing numbes of models.
Hierarchical & Sequential systems are sophisticated but hard to maintain.

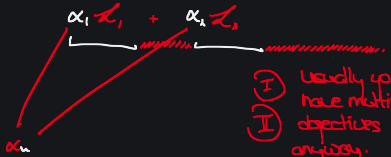


Talk is cheap. Go and try those things on MNIST to sharpen your own intuition.

The objective function defines what optimal is

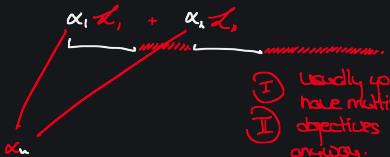


THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



Weight during inference, not during training

THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



I) Usually you have multiple objectives anyway.

II) This is a clean way to state them explicitly.

Pareto Optimization

Trivial to weight objectives while training.

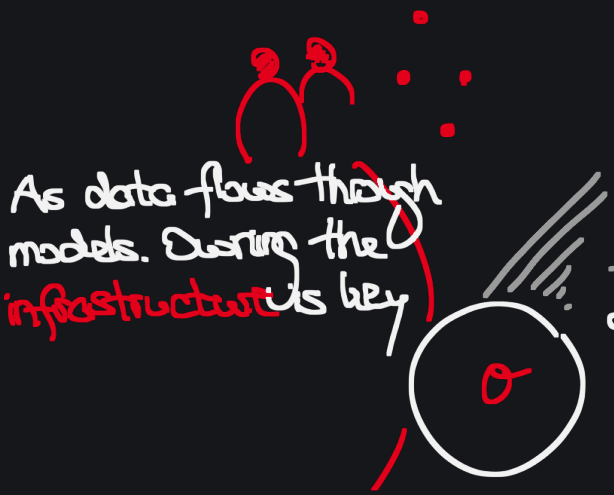
Everytime you touch is a retrain.

Beautiful opportunity to train two models and adaptively weight during inference.

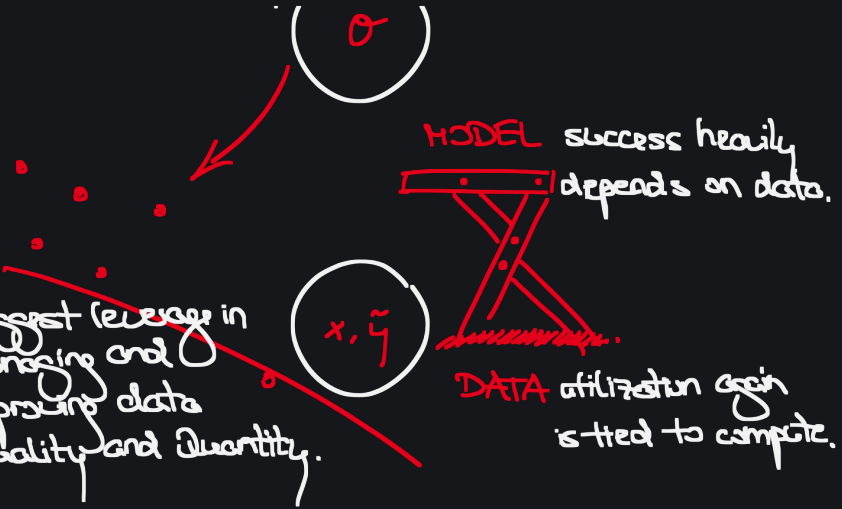
- Objective Functions require gradients.
- For everything else we still need metrics.

5

Mind Versus Data


As data flows through models. During the inference is key
Intelligent architectural designs frozen within the current paradigm.
Data-centric.

Data: model success depends on it



Better data beats better algorithms

"We don't have better algorithms. We just have better data." - Norvig (Director Search @Google)



Remember how more data is a regularization method, driving generalization of your models?

6

Summary

Summary

References

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